KAPREKAR CONTEST – FINAL - SUB JUNIOR

AMTI-NOVEMBER 1, 2003.

Instructions:
1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Attach the filled-in **FACE SLIP** and your rough working sheets along with the answer script.
5. Maximum time allowed is THREE hours.

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1. Find the quotient of the *l.c.m.* of the first 40 natural numbers divided by the *l.c.m* of the first 30 natural numbers.

2. Vasanth read an eleven-pages article in an Encyclopedia. The sum of the page number, which he read, is 1 less than 2003. What are the eleven page numbers?

3. Natural numbers are written in a sequence as follows:

   1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 ....

   What is the 2003^{rd} digit in this sequence?

4. Place one non-zero digit in each box given below, in such a way that the resulting equation is valid:

   $$\boxed{}\boxed{}\ \%\ \text{of}\ \boxed{}\boxed{}\boxed{} = 400.$$

5. Let us assign natural number values to English alphabets as follows:

   $A=1, B=2, C=3, D=4, E=5, \ldots\ldots\ldots\ldots, Z=26.$

   Let us define the "Value" of a word as the "product of the numbers represented by the letters in that word. (For example, the "Value" of the word *CAT* is $3 \times 1 \times 20 = 60$). Find an English word whose "Value" is 455.

6. In a Sports meet, one sportsman told another. "There are nine fewer of us Here, than twice the product of our total number". How many Sportsmen were there at the Meet?

7. Using only the digits 0,1,2,3,5 (with no repetitions), three-digit numbers are formed. How many of them are multiples of 6?

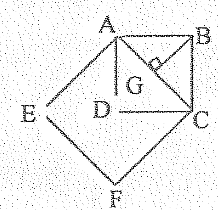
8. The squares of two consecutive positive integers differ by 2003. What is the sum of these integers?

9. What is the smallest natural number n greater than 1, such that $\sqrt{1+2+3+.......+n}$ is a positive integer?

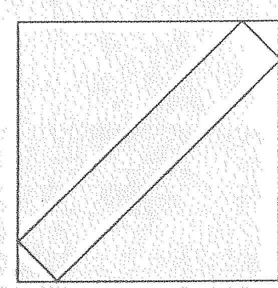
10. 42 identical cubes, each with $1cm$ edge are glued together to form a cuboid. If the perimeter of the perimeter of the base of the cuboid is $18cm.$, find the height of the cuboid.

11. The sizes of the copier paper have the property that a sheet of paper cut in half gives two smaller sheets of the same shape as the original sheet. Find the ratio of the sides of the sheets.

12. In the figure, $ABCD$ and $AEFC$ are squares. Δ ABG is a right triangle. If Δ ABG has an area $\frac{1}{4}$, find the area of the polygon $AEFCD$.



13. Two identical isosceles right triangles are removed from each opposite corner of a square resulting in a rectangle. If the sum of the areas of the cut-off pieces is 450, find the length of the diagonal of the rectangle.



14. A circle is inscribed in a rhombus, one of whose angles is 60^0. Find the ratio of the area of the rhombus to the area of the inscribed circle.

15. Ramanujan went out of his house between 2 and 3 and returning home between 7 and 8, found that the hands of the clock at his residence had exactly changed places. Find the time of his departure.

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